

Laboratory Record

submitted as part of

Programming for Data Science Lab (MACSE502)

LAB ASSESSMENT-7

by



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To



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School of Computer Science and Engineering



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

AUGUST 2026

2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record

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Week No	Date	Sl. No.	Type	Topic	Title with download Link for document	Problem Statement	Page No.
1	09-07-2026	1	PP	Pandas	Basic Data Manipulation with Pandas	Create a DataFrame with columns Name, Age, and City containing data for five individuals. Perform the following operations: select only the Name and Age columns, filter rows where Age is greater than 25, add a new column Country with a default value, and sort the DataFrame by Age in descending order. Finally, calculate the average age of the individuals.	1
1	09-07-2026	2	EP	Pandas	AI Model Performance Analysis Using Pandas	An AI company maintains model performance records containing: - ModelID - Accuracy - Algorithm Perform: - Select ModelID and Accuracy. - Filter models with Accuracy greater than 90%. - Add Status with default value "Production Ready". - Sort models by Accuracy. - Calculate average model accuracy.	6
2	16-07-2026	3	PP	Pandas	Data Cleaning and Preprocessing with Pandas	Create a DataFrame with some missing values in columns Name, Age, and City. Perform the following operations: fill missing values in the Age column with the mean age, drop rows where Name or City is missing, and convert the Age column to integer type. Finally, normalize the Age column using min-max scaling	14
2	16-07-2026	4	EP	Pandas	Movie Rating Data Preprocessing Using Pandas	A streaming platform maintains movie records containing: - MovieID - Rating - Genre Perform: - Create a dataset with missing values. - Fill missing Rating values using average rating. - Remove rows with missing MovieID or Genre. - Convert Rating into integer type. Normalize rating values.	18

3	23-07-2026	5	PP	NumPy	Creating and Manipulating Arrays Using NumPy	Create a 2D array of shape (3, 4) with random integers between 0 and 10. Perform the following operations: calculate the mean and standard deviation of the entire array, slice the array to get the first two rows and last two columns, reshape the array to shape (4, 3), and perform element-wise multiplication with another array of the same shape.	30
3	23-07-2026	6	EP	NumPy	Warehouse Shipment and Logistic Analytics Using Numpy	A logistics company tracks shipments from 4 warehouses across 7 weeks. Create a 4 x 7 shipment matrix. Perform the following operations: (1) compute average shipments, (2) compute standard deviation, (3) slice the first two warehouses and last three weeks, (4) reshape the matrix into a 7 x 4 matrix, and (5) perform element-wise multiplication with a transportation-cost matrix of the same shape. As an extra	35
4	30/7/2026	7	PP	Pandas, Numpy, Matplotlib	Feature Engineering and Exploratory Data Analysis	Create a DataFrame with columns Feature1, Feature2, and Target containing random data. Perform the following operations: create a new feature that is the logarithm of Feature1, bin Feature2 into three categories (low, medium, high), and calculate the correlation matrix of the DataFrame. Finally, create a scatter plot of Feature1 vs. Feature2 colored by Target, and interpret any visible patterns.	46

4	30/7/2026	8	EP	Pandas, NumPy, Matplotlib	Employee Attrition Prediction Using Workforce Indicators	A company wants to analyze workforce behavior and predict employee attrition. Create a DataFrame containing: OvertimeHours JobSatisfactionScore SalaryLevel TrainingHours WorkLifeBalance YearsAtCompany PromotionCount AttritionStatus Perform: 1. Create Log_OvertimeHours. 2. Categorize JobSatisfactionScore into: - o Disengaged - o Satisfied - o Highly Engaged 3. Generate correlation matrix. 4. Create a Treemap Visualization of employee categories. Analyze workforce trends.	54
5	06/08/2026	9	PP	Web Scraping	requests, BeautifulSoup, pandas, json	Scrape data from a webpage and store it in a structured format like CSV or JSON	63
5	06/08/2026	10	EP	Web Scraping	Online Job Market Analytics	A recruitment agency wants to analyze job opportunities posted on a dynamic job portal. Develop a Python application using Selenium to scrape job listings, extract Job Title, Company Name, Location and Salary (if available), store the collected data in a Pandas DataFrame, save the dataset as a CSV file, and visualize job demand by location.	70
6	20/08/2026	11	PP	Plotly	Interactive Bar Chart with Plotly	Create an interactive bar chart showing the population of different cities.	80
6	20/08/2026	12	PP	Plotly	Interactive Scatter Plot with Plotly	Create an interactive scatter plot showing the relationship between house size and price.	88

7	27/08/2026	13	PP	Bokeh	Interactive Scatter Plot with Bokeh	Create an interactive scatter plot showing the relationship between petal length and petal width from the Iris dataset using Bokeh.	101
7	27/08/2026	14	EP	Bokeh	Interactive Line Plot with Bokeh	Create an interactive line plot showing the daily temperatures over a week using Bokeh.	105
7	27/08/2026	15	EP	Bokeh	Interactive Bar Chart with Bokeh	Create an interactive bar chart showing the sales figures of different products using Bokeh.	109

PP: Practice Program; EP: Exercise Program; CP: Challenging Program

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Practice Program

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Program S. No. 9

Title	Web Scraping and Data Collection			
Problem Statement	Scrape data from a webpage and store it in a structured format like CSV or JSON format			
Objectives	• To understand the concept of web scraping and data collection. • To retrieve webpage content using the Requests library. • To extract required information from HTML using BeautifulSoup. • To organize the scraped data using Pandas DataFrame. • To store the collected data in structured formats such as CSV and JSON.			
Dataset URL Link / File	https://quotes.toscrape.com			
Function Details	Sl. No.	Function	Package	Description of Function
	1	requests.get()	requests	Retrieves the HTML content of a webpage
	2	BeautifulSoup()	bs4	Parses the HTML content of the webpage
	3	find_all()	BeautifulSoup	Finds all matching HTML elements
	4	find()	BeautifulSoup	Finds the first matching HTML element
	5	DataFrame()	pandas	Converts collected data into tabular format
	6	to_csv()	pandas	Stores data in CSV format
	7	json.dump()	json	Stores Python data in JSON format
	8	json.load()	json	Reads data from a JSON file
Steps/ Process/ Procedure/	1. Import the required libraries: requests, BeautifulSoup, pandas and json. 2. Specify the URL of the webpage to be scraped.			

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Algorithm/ Flowchart/ Model	3. Send an HTTP GET request using requests.get(). 4. Parse the HTML content using BeautifulSoup. 5. Locate the required HTML elements containing quotes, authors and tags. 6. Extract the required data from the webpage. 7. Store the extracted data in a Python list. 8. Convert the collected data into a Pandas DataFrame. 9. Save the structured data in CSV format using to_csv(). 10. Save the same data in JSON format using json.dump(). 11. Display the collected data to verify the output.
Program	<pre> import requests from bs4 import BeautifulSoup import pandas as pd import json url = "https://quotes.toscrape.com/" response = requests.get(url) print("Status Code:", response.status_code) soup = BeautifulSoup(response.text, "html.parser") print("Website Title:", soup.title.text) quotes = [] for item in soup.find_all("div", class_="quote"): quote = item.find("span", class_="text").text author = item.find("small", class_="author").text tags = [tag.text for tag in item.find_all("a", class_="tag")] quotes.append({ "Quote": quote, "Author": author, "Tags": " ".join(tags) }) print("Number of quotes collected:", len(quotes)) </pre>

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```
df = pd.DataFrame(quotes)

df

df.to_csv("quotes.csv", index=False)

print("Data saved successfully in CSV format.")

with open("quotes.json", "w", encoding="utf-8") as file:
    json.dump(quotes, file, indent=4, ensure_ascii=False)

print("Data saved successfully in JSON format.")

csv_data = pd.read_csv("quotes.csv")

csv_data

all_quotes = []

page = 1

while True:

    url = f"https://quotes.toscrape.com/page/{page}/"

    response = requests.get(url)
    soup = BeautifulSoup(response.text, "html.parser")

    items = soup.find_all("div", class_="quote")

    if not items:
        break

    for item in items:

        quote = item.find("span", class_="text").text
        author = item.find("small", class_="author").text

        tags = [
            tag.text
            for tag in item.find_all("a", class_="tag")
        ]

        all_quotes.append({
            "Quote": quote,
```

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	<pre> "Author": author, "Tags": ", ".join(tags) }) page += 1 print("Total quotes collected:", len(all_quotes)) df = pd.DataFrame(all_quotes) df df.to_csv("quotes_dataset.csv", index=False) with open("quotes_dataset.json", "w", encoding="utf-8") as file: json.dump(all_quotes, file, indent=4, ensure_ascii=False) print("CSV and JSON files created successfully.") </pre>
Output Snapshots With Explanation/ Observations	The required libraries requests, BeautifulSoup, pandas and json are imported for web scraping, HTML parsing, data organization and structured data storage. <pre> 1 import requests 2 from bs4 import BeautifulSoup 3 import pandas as pd 4 import json 1 url = "https://quotes.toscrape.com/" 2 3 response = requests.get(url) 4 5 print("Status Code:", response.status_code) </pre> Status Code: 200 <i>Fig. 1.1: Importing Required Python Libraries</i>

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The webpage <https://quotes.toscrape.com/> is accessed using `requests.get()`. The status code 200 confirms that the webpage was successfully retrieved.

```
[7]: 1 quotes = []
      2
      3 for item in soup.find_all("div", class_="quote"):
      4
      5     quote = item.find("span", class_="text").text
      6     author = item.find("small", class_="author").text
      7
      8     tags = [
      9         tag.text
     10         for tag in item.find_all("a", class_="tag")
     11     ]
     12
     13 quotes.append({
     14     "Quote": quote,
     15     "Author": author,
     16     "Tags": ", ".join(tags)
     17 })
     18
     19 print("Number of quotes collected:", len(quotes))

Number of quotes collected: 10
```

Fig. 1.2: Accessing the Webpage

The HTML content of the webpage is parsed using BeautifulSoup. The webpage title confirms that the correct page has been successfully loaded.

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```
number of quotes collected: 10

[8]: 1 df = pd.DataFrame(quotes)
      2
      3 df

[8]:
```

	Quote	Author	Tags
0	"The world as we have created it is a process ...	Albert Einstein	change, deep-thoughts, thinking, world
1	"It is our choices, Harry, that show what we t...	J.K. Rowling	abilities, choices
2	"There are only two ways to live your life. On...	Albert Einstein	inspirational, life, live, miracle, miracles
3	"The person, be it gentleman or lady, who has ...	Jane Austen	aliteracy, books, classic, humor
4	"Imperfection is beauty, madness is genius and...	Marilyn Monroe	be-yourself, inspirational
5	"Try not to become a man of success. Rather be...	Albert Einstein	adulthood, success, value
6	"It is better to be hated for what you are tha...	André Gide	life, love
7	"I have not failed. I've just found 10,000 way...	Thomas A. Edison	edison, failure, inspirational, paraphrased
8	"A woman is like a tea bag; you never know how...	Eleanor Roosevelt	misattributed-eleanor-roosevelt
9	"A day without sunshine is like, you know, nig...	Steve Martin	humor, obvious, simile

Fig. 1.3: Parsing the HTML and displaying the dataset Content

```
1 df.to_csv("quotes.csv", index=False)
2
3 print("Data saved successfully in CSV format.")

Data saved successfully in CSV format.
```

Fig. 1.4: Saving Data in CSV Format

The scraped data is successfully stored in CSV format using the pandas `to_csv()` function.

The scraped data is successfully stored in JSON format using the `json.dump()` function.

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```
1 with open("quotes.json", "w", encoding="utf-8") as file:
2     json.dump(quotes, file, indent=4, ensure_ascii=False)
3
4 print("Data saved successfully in JSON format.")
```

Data saved successfully in JSON format.

Fig. 1.5: Saving Data in JSON Format

	Quote	Author	Tags
0	"The world as we have created it is a process ...	Albert Einstein	change, deep-thoughts, thinking, world
1	"It is our choices, Harry, that show what we t...	J.K. Rowling	abilities, choices
2	"There are only two ways to live your life. On...	Albert Einstein	inspirational, life, live, miracle, miracles
3	"The person, be it gentleman or lady, who has ...	Jane Austen	aliteracy, books, classic, humor
4	"Imperfection is beauty, madness is genius and...	Marilyn Monroe	be-yourself, inspirational
5	"Try not to become a man of success. Rather be...	Albert Einstein	adulthood, success, value
6	"It is better to be hated for what you are tha...	André Gide	life, love
7	"I have not failed. I've just found 10,000 way...	Thomas A. Edison	edison, failure, inspirational, paraphrased
8	"A woman is like a tea bag; you never know how...	Eleanor Roosevelt	misattributed-eleanor-roosevelt
9	"A day without sunshine is like, you know, nig...	Steve Martin	humor, obvious, simile

Fig 1.6: Displaying the saved json and csv

The scraped data is successfully stored in JSON format using the `json.dump()` function.

The saved CSV file is read using `pandas.read_csv()` and displayed to verify that the data has been stored correctly.

The saved JSON file is read using `json.load()` and the stored records are displayed successfully, confirming correct JSON storage.

Conclusions

The experiment was successfully completed by scraping data from the given webpage using Python. The webpage content was retrieved using the requests library and parsed using BeautifulSoup. The required information such as quotes, authors, and tags was extracted and organized using a Pandas DataFrame. Finally, the collected data was successfully stored in structured CSV and JSON formats for further analysis and use.

References

1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data
2. Python Program: [GitHub link](#)
3. Class lecture: [Lecture link](#)

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Program S. No. 10

Title	Online Job Market Analytics			
Problem Statement	A recruitment agency wants to analyze job opportunities posted on a dynamic job portal. Develop a Python application using Selenium to scrape job listings, extract Job Title, Company Name, Location and Salary (if available), store the collected data in a Pandas DataFrame, save the dataset as a CSV file, and visualize job demand by location.			
Objectives	• To understand web scraping from a dynamic job portal using Selenium. • To extract job title, company name, location and salary information from job listings. • To organize and clean the collected data using a Pandas DataFrame. • To save the structured job dataset in CSV format. • To perform location-wise analysis and visualize job distribution using Matplotlib.			
Dataset URL Link / File	Job Portal: https://www.naukri.com Search role used: Data Scientist Cities: Bangalore, Hyderabad, Pune, Chennai, Mumbai and Delhi			
Functions Details	Sl. No.	Function	Package	Description of function
	1	Firefox()	selenium	Starts a Firefox browser session controlled by Selenium.
	2	driver.get()	selenium	Opens the specified job-portal URL in the automated browser.
	3	execute_script()	selenium	Executes JavaScript to scroll the dynamically loaded webpage.
	4	find_elements()	selenium	Finds all matching job-card elements on the page.
	5	find_element()	selenium	Finds a required field inside an individual job card.
	6	time.sleep()	time	Pauses execution briefly to allow dynamic content to load.
	7	DataFrame()	pandas	Converts the collected job records into tabular form.
	8	drop_duplicates()	pandas	Removes duplicate job records from the DataFrame.
	9	reset_index()	pandas	Resets row indexing after data cleaning.
	10	to_csv()	pandas	Saves the structured job dataset as a CSV file.
	11	value_counts()	pandas	Counts job records for each location for analysis.
	12	bar()/savefig()	matplotlib	Creates and saves the location-wise job-demand bar chart.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Selenium, Pandas, Matplotlib and supporting Python modules. 2. Define the job role, location and maximum number of job listings to collect. 3. Launch the browser using Selenium WebDriver and construct the Naukri search URL. 4. Open the dynamic job-search page and allow sufficient time for the page to load. 5. Scroll the webpage so that dynamically loaded job cards become available. 6. Locate job-card elements using CSS selectors. 7. Extract Job Title, Company Name, Location and Salary from each job card. 8. Store the extracted records in a Python list and convert them into a Pandas DataFrame. 9. Remove duplicate records and save the cleaned dataset as a CSV file. 10. Repeat the search for Bangalore, Hyderabad, Pune, Chennai, Mumbai			

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	and Delhi. 11. Use value_counts() to prepare location-wise job analysis and create a bar chart. 12. Save the visualization and multi-city CSV file, display the final summary and close the browser. START -> Open Job Portal -> Scrape Listings -> DataFrame -> CSV -> Location Analysis -> Bar Chart -> STOP
Program	<pre> import time import pandas as pd import matplotlib.pyplot as plt from selenium import webdriver from selenium.webdriver.common.by import By from selenium.webdriver.firefox.options import Options from selenium.common.exceptions import NoSuchElementException JOB_ROLE = "data scientist" LOCATION = "bangalore" MAX_JOBS = 30 options = Options() driver = webdriver.Firefox(options=options) driver.set_window_size(1400, 900) def make_slug(text): return "-".join(text.lower().strip().split()) def get_text(card, selectors, default="Not Available"): for selector in selectors: try: element = card.find_element(By.CSS_SELECTOR, selector) text = element.text.strip() if text: return text except NoSuchElementException: pass return default job_slug = make_slug(JOB_ROLE) location_slug = make_slug(LOCATION) url = f"https://www.naukri.com/{job_slug}-jobs-in-{location_slug}" print("Search URL:", url) driver.get(url) time.sleep(5) print("Page Title:", driver.title) print("Current URL:", driver.current_url) </pre>

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```
for i in range(3):
    driver.execute_script("window.scrollTo(0,
document.body.scrollHeight);")
    time.sleep(2)
driver.execute_script("window.scrollTo(0, 0);")

cards = driver.find_elements(By.CSS_SELECTOR, "div.cust-job-tuple")
print("Number of job cards found:", len(cards))

jobs = []
for card in cards[:MAX_JOBS]:
    title = get_text(card, ["a.title", "[class*='title']"])
    company = get_text(card, ["a.comp-name", ".comp-name"])
    location = get_text(card, ["span.locWdth", ".locWdth",
"[class*='location']"])
    salary = get_text(card, ["span.sal", ".sal", "[class*='salary']"],
default="Not Disclosed")
    jobs.append({
        "Job Title": title,
        "Company Name": company,
        "Location": location,
        "Salary": salary
    })

print("Jobs scraped:", len(jobs))
df = pd.DataFrame(jobs)
df = df.drop_duplicates().reset_index(drop=True)
print("Total unique jobs:", len(df))
print("Dataset Shape:", df.shape)
print(df.head(10))
print("Salary Information:")
print(df["Salary"].value_counts())

df.to_csv("job_market_data.csv", index=False)
print("CSV file saved successfully.")

csv_data = pd.read_csv("job_market_data.csv")
print(csv_data.head())

cities = ["bangalore", "hyderabad", "pune", "chennai", "mumbai",
"delhi"]
all_jobs = []

for city in cities:
    city_slug = make_slug(city)
    url = f"https://www.naukri.com/{job_slug}-jobs-in-{city_slug}"
    print("Scraping:", city)
```

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```
driver.get(url)

time.sleep(4)

city_cards = driver.find_elements(By.CSS_SELECTOR, "div.cust-job-  
tuple")
print("Jobs found:", len(city_cards))

for card in city_cards[:10]:
    title = get_text(card, ["a.title", "[class*='title']"])
    company = get_text(card, ["a.comp-name", ".comp-name"])
    salary = get_text(card, ["span.sal", ".sal", "[class*='salary']"],  
                        default="Not Disclosed")
    all_jobs.append({
        "Job Title": title,
        "Company Name": company,
        "Location": city.title(),
        "Salary": salary
    })

city_df = pd.DataFrame(all_jobs)
city_df = city_df.drop_duplicates().reset_index(drop=True)
print("Total jobs collected:", len(city_df))

location_analysis = city_df["Location"].value_counts().reset_index()
location_analysis.columns = ["Location", "Number of Jobs"]
print(location_analysis)

plt.figure(figsize=(10, 5))
plt.bar(location_analysis["Location"], location_analysis["Number of  
Jobs"])
plt.title("Data Scientist Job Demand by Location")
plt.xlabel("Location")
plt.ylabel("Number of Job Listings")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("job_demand_by_location.png", dpi=300,  
bbox_inches="tight")
plt.show()

city_df.to_csv("job_market_multi_city.csv", index=False)
print("Multi-city dataset saved successfully.")

print("ONLINE JOB MARKET ANALYTICS")
print("Job Role:", JOB_ROLE)
print("Total Jobs Analyzed:", len(city_df))
print("Cities Analyzed:", len(location_analysis))
```

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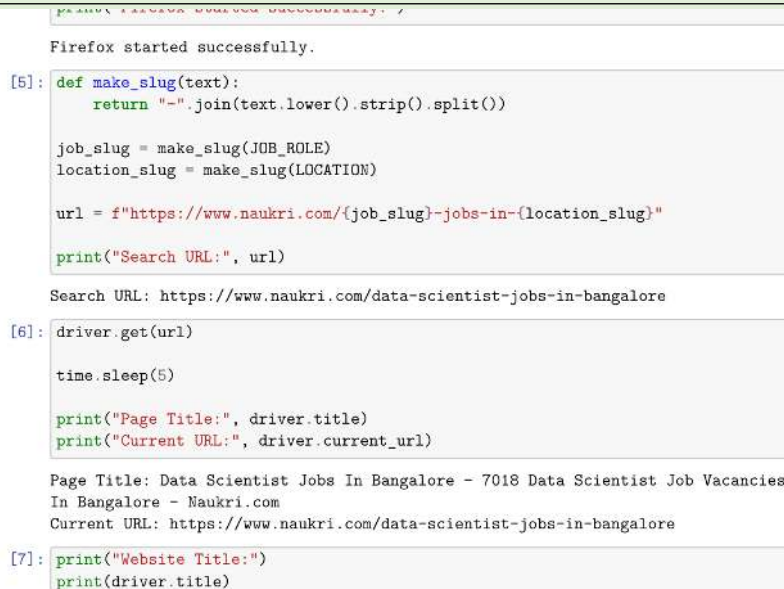
Week No.: 5

Program S. No. 10

**Output
Snapshots
With
Explanation/
Observations**

```
print(location_analysis.to_string(index=False))

driver.quit()
print("Browser closed successfully.")
```



```

# ===== OUTPUT =====
Firefox started successfully.

[5]: def make_slug(text):
      return "-".join(text.lower().strip().split())

      job_slug = make_slug(JOB_ROLE)
      location_slug = make_slug(LOCATION)

      url = f"https://www.naukri.com/{job_slug}-jobs-in-{location_slug}"

      print("Search URL:", url)

Search URL: https://www.naukri.com/data-scientist-jobs-in-bangalore

[6]: driver.get(url)

      time.sleep(5)

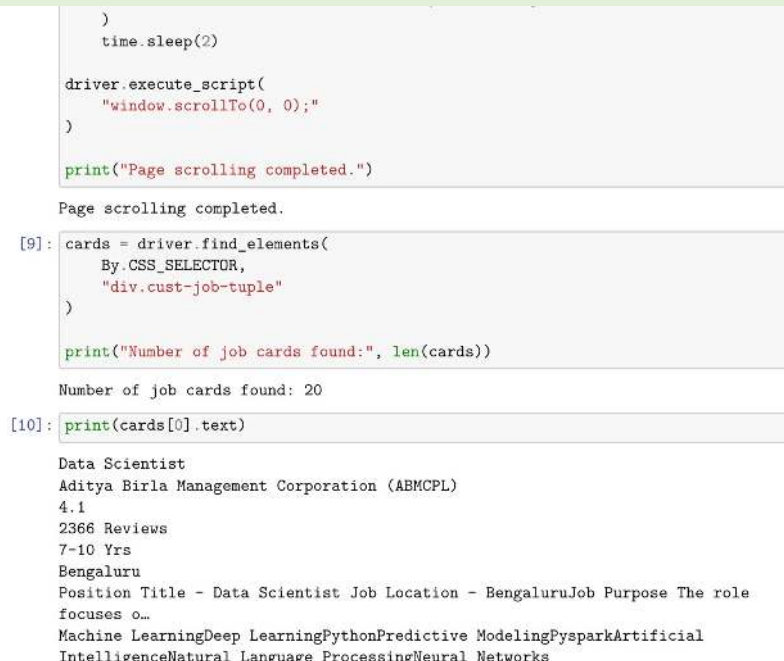
      print("Page Title:", driver.title)
      print("Current URL:", driver.current_url)

Page Title: Data Scientist Jobs In Bangalore - 7018 Data Scientist Job Vacancies
In Bangalore - Naukri.com
Current URL: https://www.naukri.com/data-scientist-jobs-in-bangalore

[7]: print("Website Title:")
      print(driver.title)
  
```

Fig. 1.1: Search URL and Job Portal Validation

The Selenium session generated the Naukri search URL for Data Scientist jobs in Bangalore and successfully retrieved the page title and current URL, confirming that the dynamic job portal was opened correctly.



```

      )
      time.sleep(2)

      driver.execute_script(
          "window.scrollTo(0, 0);"
      )

      print("Page scrolling completed.")

Page scrolling completed.

[9]: cards = driver.find_elements(
      By.CSS_SELECTOR,
      "div.cust-job-tuple"
    )

      print("Number of job cards found:", len(cards))

Number of job cards found: 20

[10]: print(cards[0].text)

Data Scientist
Aditya Birla Management Corporation (ABMCPL)
4.1
2366 Reviews
7-10 Yrs
Bengaluru
Position Title - Data Scientist Job Location - BengaluruJob Purpose The role
focuses o...
Machine LearningDeep LearningPythonPredictive ModelingPysparkArtificial
IntelligenceNatural Language ProcessingNeural Networks
  
```

Fig. 1.2: Detection of Dynamic Job Cards

After scrolling the page, Selenium detected 20 job-card elements. The

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first job card displays details such as job title, company, experience, location and associated skills.

```
jobs.append({
    "Job Title": title,
    "Company Name": company,
    "Location": location,
    "Salary": salary
})

print("Jobs scraped:", len(jobs))

Jobs scraped: 20

[14]: jobs[:3]

[14]: [{'Job Title': 'Data Scientist',
      'Company Name': 'Aditya Birla Management Corporation (ABMCPL)',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'},
      {'Job Title': 'Data Scientist/Software Developer',
      'Company Name': 'Xylem Water Solutions',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'},
      {'Job Title': 'Data Scientist III',
      'Company Name': 'Walmart',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'}]

[15]: df = pd.DataFrame(jobs)

df

[15]:
```

	Job Title \
0	Data Scientist
1	Data Scientist/Software Developer
2	Data Scientist III
3	Data Scientist
4	Data Scientist Python GenAI LLM RAG
5	Machine Learning Data Scientist
6	Advanced Data Scientist
7	Data Scientist
8	Data Scientist GenAI Agentic AI LLM RAG
9	Data Scientist AIML
10	Data Scientist/Python/MLOps- 4th Sep(Fri)-6+ y...
11	Data scientist III
12	Data Scientist
13	Data Scientist/ Data Analyst
14	Walk-in Data Scientist

Fig. 1.3: Scraped Job Records and Pandas DataFrame

The scraper collected 20 listings and extracted the required Job Title, Company Name, Location and Salary fields. The records were then converted into a structured Pandas DataFrame.

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```
7           Pune, Bengaluru Not Disclosed
8 Hybrid - Pune, Bengaluru, Delhi / NCR Not Disclosed
9           Hyderabad, Chennai, Bengaluru Not Disclosed

[18]: print("Dataset Shape:", df.shape)

      print("\nColumn Names:")
      print(df.columns.tolist())

      print("\nTotal Records:", len(df))

Dataset Shape: (20, 4)

Column Names:
['Job Title', 'Company Name', 'Location', 'Salary']

Total Records: 20

[19]: print("Salary Information:")

      print(df["Salary"].value_counts())

Salary Information:
Salary
Not Disclosed      16
20-30 Lacs PA       1
15-20 Lacs PA       1
14-20 Lacs PA       1
0-18 Lacs PA        1
Name: count, dtype: int64

[20]: df.to_csv(
      "job_market_data.csv",
```

Fig. 1.4: Dataset Structure, Salary Availability and CSV Creation

The initial dataset contains 20 records and 4 required columns. Salary information was unavailable for most listings, while a few postings contained salary ranges. The cleaned DataFrame was saved successfully as job_market_data.csv.

```
      index=False
    )
    print("CSV file saved successfully.")

CSV file saved successfully.

[21]: csv_data = pd.read_csv(
      "job_market_data.csv"
    )

    csv_data.head()

[21]:      Job Title \
0           Data Scientist
1  Data Scientist/Software Developer
2           Data Scientist III
3           Data Scientist
4 Data Scientist | Python | GenAI | LLM | RAG

      Company Name \
0 Aditya Birla Management Corporation (ABMCPL)
1           Xylem Water Solutions
2           Walmart
3           Genpact
4           Infosys

      Location      Salary
0      Bengaluru  Not Disclosed
1      Bengaluru  Not Disclosed
2      Bengaluru  Not Disclosed
3      Bengaluru  20-30 Lacs PA
```

Fig. 1.5: Verification of the Saved CSV Dataset

The generated CSV file was read back using pandas.read_csv(), and the displayed rows confirm that the job-listing data was stored correctly in

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structured form.

```
"Company Name": company,
"Location": city.title(),
"Salary": salary
})

Scraping: bangalore
Jobs found: 20

Scraping: hyderabad
Jobs found: 20

Scraping: pune
Jobs found: 20

Scraping: chennai
Jobs found: 20

Scraping: mumbai
Jobs found: 21

Scraping: delhi
Jobs found: 21

[25]: city_df = pd.DataFrame(all_jobs)

city_df = city_df.drop_duplicates().reset_index(drop=True)

print("Total jobs collected:", len(city_df))

city_df.head(10)

Total jobs collected: 60

[25]:
```

	Job Title \
0	Data Scientist
1	Data Scientist/Software Developer
2	Data Scientist III
3	Data Scientist
4	Data Scientist Python GenAI LLM RAG
5	Machine Learning Data Scientist
6	Advanced Data Scientist
7	Data Scientist

Fig. 1.6: Multi-City Job Data Collection

The same Data Scientist search was executed for Bangalore, Hyderabad, Pune, Chennai, Mumbai and Delhi. A controlled sample of 10 listings per city produced 60 records for location-wise analysis.

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```
4          Infosys Bangalore Not Disclosed
5          Hsbc Bangalore Not Disclosed
6          ExxonMobil Bangalore Not Disclosed
7          VOIS Bangalore Not Disclosed
8          Infosys Bangalore Not Disclosed
9          Tata Consultancy Services Bangalore Not Disclosed

[26]: location_analysis = (
      city_df["Location"]
      .value_counts()
      .reset_index()
      )

      location_analysis.columns = [
          "Location",
          "Number of Jobs"
      ]

      location_analysis

[26]:   Location  Number of Jobs
0  Bangalore           10
1  Hyderabad           10
2    Pune           10
3  Chennai           10
4   Mumbai           10
5    Delhi           10

[27]: plt.figure(figsize=(10, 5))
```

Fig. 1.7: Location-Wise Job Analysis

Pandas value_counts() produced the location-wise summary for the six-city job sample.

```
plt.tight_layout()
plt.show()
```

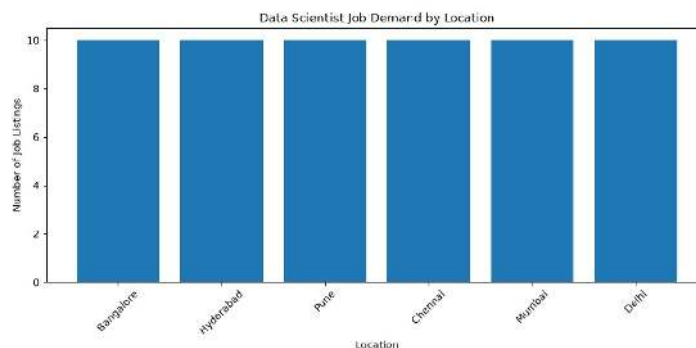


Fig. 1.8: Data Scientist Job Demand Visualization by Location

A Matplotlib bar chart visualizes the number of sampled job listings for each selected city. The graph provides a clear location-wise comparison of the collected job-market sample.

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```
[30]: print("ONLINE JOB MARKET ANALYTICS")
      print("-" * 35)

      print("Job Role:", JOB_ROLE)
      print("Total Jobs Analyzed:", len(city_df))
      print("Cities Analyzed:", len(location_analysis))

      print("\nLocation-wise Job Demand:")
      print(location_analysis.to_string(index=False))

ONLINE JOB MARKET ANALYTICS
-----
Job Role: data scientist
Total Jobs Analyzed: 60
Cities Analyzed: 6

Location-wise Job Demand:
Location  Number of Jobs
Bangalore           10

Hyderabad           10
Pune                 10
Chennai              10
Mumbai               10
Delhi                 10

[31]: driver.quit()

print("Browser closed successfully")
```

Fig. 1.9: Final Job Market Analytics Summary

The final output confirms that the multi-city dataset and graph were saved successfully. The analysis includes 60 sampled job records across 6 cities and displays the location-wise summary.

Conclusions

The exercise successfully demonstrated Selenium-based web scraping on a dynamic job portal. Job title, company name, location and available salary information were extracted from Naukri, organized in Pandas DataFrames, cleaned and stored as CSV files. Location-wise analysis was performed for six cities and visualized using a Matplotlib bar chart. Thus, the required job listings dataset, CSV output, location analysis and demand visualization were generated successfully.

References

1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018.
2. Python Program (GitHub link): <https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/Lab%20Codes/Week5-EP.ipynb>
3. Class lecture link: <https://drive.google.com/file/d/1HjCacfaFs49XEnuM3idduuKB0Tgjb4xI/view>

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Title	Interactive Bar Chart with Plotly			
Problem Statement	Create an interactive bar chart showing the population of different cities.			
Objectives	• To understand the creation of interactive data visualizations using Plotly. • To organize city population data in a Pandas DataFrame. • To create an interactive bar chart for comparing population across different cities. • To sort and analyze population data to identify the highest, lowest, and average values. • To save the interactive chart as an HTML file and the dataset as a CSV file.			
Dataset URL Link / File	City		Population (Lakhs)	
	Mumbai		204	
	Delhi		167	
	Bengaluru		136	
	Hyderabad		108	
	Chennai		115	
	Kolkata		149	
	Dataset source: Manually defined in the Jupyter Notebook for this exercise; no external dataset URL is required.			
Functions Details	Sl. No.	Function	Package	Description of function
	1	pd.DataFrame()	pandas	Creates a structured DataFrame from the city and population data.
	2	px.bar()	plotly.express	Creates an interactive bar chart from the DataFrame.
	3	update_traces()	plotly	Updates trace properties such as the position of value labels on the bars.
	4	update_layout()	plotly	Updates chart layout properties including axis titles.
	5	show()	plotly	Displays the interactive Plotly figure in the Jupyter Notebook.
	6	sort_values()	pandas	Sorts the DataFrame based on population values in ascending or descending order.
	7	idxmax()	pandas	Returns the index of the row containing

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				the maximum population value.
	8	idxmin()	pandas	Returns the index of the row containing the minimum population value.
	9	mean()	pandas	Calculates the average population of all cities.
	10	round()	Python	Rounds the calculated average population to the required decimal places.
	11	write_html()	plotly	Saves the interactive Plotly chart as an HTML file.
	12	to_csv()	pandas	Saves the DataFrame as a CSV file without the index column.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import the Pandas library and Plotly Express for data handling and interactive visualization. 2. Create a dictionary containing the names of six Indian cities and their population values in lakhs. 3. Convert the dictionary into a Pandas DataFrame and display the dataset. 4. Check the dataset shape and print the city and population lists to verify the input data. 5. Create an interactive bar chart using Plotly Express with City on the X-axis and Population (Lakhs) on the Y-axis. 6. Display population values above the bars and provide appropriate chart and axis titles. 7. Sort the DataFrame in descending order of population using <code>sort_values()</code>. 8. Create and display a second interactive bar chart using the sorted population data. 9. Use <code>idxmax()</code> and <code>idxmin()</code> to identify the cities with the highest and lowest populations. 10. Calculate the average population using <code>mean()</code> and round the result to two decimal places. 11. Display the final population analysis summary in the notebook. 12. Save the interactive Plotly chart as <code>city_population_chart.html</code> and save the dataset as <code>city_population.csv</code>. Flow: START → Import Libraries → Create Dataset → DataFrame → Verify Data → Interactive Bar Chart → Sort Data → Sorted Chart → Population Statistics → Save HTML/CSV → STOP			
Program	import pandas as pd			

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```
import plotly.express as px

data = {
    "City": [
        "Mumbai",
        "Delhi",
        "Bengaluru",
        "Hyderabad",
        "Chennai",
        "Kolkata"
    ],
    "Population (Lakhs)": [
        204,
        167,
        136,
        108,
        115,
        149
    ]
}

df = pd.DataFrame(data)
df

print("Dataset Shape:", df.shape)
print("\nCities:")
print(df["City"].tolist())
print("\nPopulation:")
print(df["Population (Lakhs)"].tolist())

fig = px.bar(
    df,
    x="City",
    y="Population (Lakhs)",
    title="Population of Major Indian Cities",
    text="Population (Lakhs)"
)
fig.update_traces(textposition="outside")
fig.update_layout(
    xaxis_title="City",
    yaxis_title="Population (Lakhs)"
)
fig.show()

sorted_df = df.sort_values(
    by="Population (Lakhs)",
```

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```
        ascending=False
    )
    sorted_df

    fig = px.bar(
        sorted_df,
        x="City",
        y="Population (Lakhs)",
        title="City-wise Population Analysis",
        text="Population (Lakhs)"
    )
    fig.update_traces(textposition="outside")
    fig.update_layout(
        xaxis_title="City",
        yaxis_title="Population (Lakhs)"
    )
    fig.show()

    highest = df.loc[
        df["Population (Lakhs)"].idxmax()
    ]
    print("City with Highest Population:")
    print(highest)

    lowest = df.loc[
        df["Population (Lakhs)"].idxmin()
    ]
    print(
        "Lowest Population:",
        lowest["City"],
        "-",
        lowest["Population (Lakhs)"],
        "Lakhs"
    )

    average_population = df["Population (Lakhs)"].mean()
    print(
        "Average Population:",
        round(average_population, 2),
        "Lakhs"
    )

    print("CITY POPULATION ANALYSIS")
    print("-" * 35)
    print("Total Cities:", len(df))
    print(
```

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```
"Highest Population:",
df.loc[df["Population (Lakhs)"].idxmax(), "City"]
)
print(
    "Lowest Population:",
    df.loc[df["Population (Lakhs)"].idxmin(), "City"]
)
print(
    "Average Population:",
    round(df["Population (Lakhs)"].mean(), 2),
    "Lakhs"
)

fig.write_html(
    "city_population_chart.html"
)
print("Interactive chart saved successfully.")

df.to_csv(
    "city_population.csv",
    index=False
)
print("Dataset saved successfully.")
```

**Output
Snapshots
With
Explanation/
Observations**

The following output snapshots are taken from the completed Jupyter Notebook. Each image is inserted at full content width so that the dataset, charts, code cells, and output values remain readable and are not cut off.

```
[5]:      City  Population (Lakhs)
0    Mumbai             204
1     Delhi             167
2  Bengaluru             136
3   Hyderabad             108
4     Chennai             115
5    Kolkata             149

[6]: print("Dataset Shape:", df.shape)

print("\nCities:")
print(df["City"].tolist())

print("\nPopulation:")
print(df["Population (Lakhs)"].tolist())

Dataset Shape: (6, 2)

Cities:
['Mumbai', 'Delhi', 'Bengaluru', 'Hyderabad', 'Chennai', 'Kolkata']

Population:
[204, 167, 136, 108, 115, 149]

[8]: fig = px.bar(
      df,
```

Fig. 1.1: Population Dataset and Dataset Structure

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Explanation/Observation: The DataFrame verifies 6 cities and their corresponding population values.

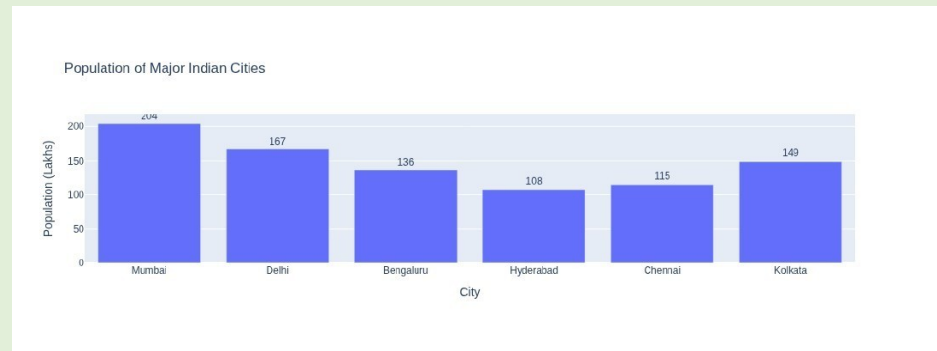


Fig. 1.2: Interactive Bar Chart of Major Indian Cities

Explanation/Observation: Plotly Express generates an interactive bar chart for the original city order. The population values are displayed above the bars, with Mumbai showing the highest value of 204 lakhs in the given dataset.

```
[9]: sorted_df = df.sort_values(
    by="Population (Lakhs)",
    ascending=False
)

sorted_df
```

	City	Population (Lakhs)
0	Mumbai	204
1	Delhi	167
5	Kolkata	149
2	Bengaluru	136
4	Chennai	115
3	Hyderabad	108

Fig. 1.3: Population Data Sorted in Descending Order

Explanation/Observation: The DataFrame is sorted using `sort_values()` in descending order of population. The resulting order is Mumbai, Delhi, Kolkata, Bengaluru, Chennai, and Hyderabad.



Fig. 1.4: City-wise Population Analysis after Sorting

Explanation/Observation: The sorted bar chart displays the cities from

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highest to lowest population, making comparison and ranking easier.

```
[11]: highest = df.loc[
      df["Population (Lakhs)"].idxmax()
    ]

    print("City with Highest Population:")
    print(highest)
```

```
City with Highest Population:
City                Mumbai
Population (Lakhs)    204
Name: 0, dtype: object
```

```
[12]: lowest = df.loc[
      df["Population (Lakhs)"].idxmin()
    ]

    print(
        "Lowest Population:",
        lowest["City"],
        "-",
        lowest["Population (Lakhs)"],
        "Lakhs"
    )
```

```
Lowest Population: Hyderabad - 108 Lakhs
```

```
[13]: average_population = df["Population (Lakhs)"].mean()

    print(
        "Average Population:",
        round(average_population, 2),
        "Lakhs"
    )
```

```
Average Population: 146.5 Lakhs
```

Fig. 1.5: Highest, Lowest, and Average Population Analysis

Explanation/Observation: The program identifies Mumbai as the city with the highest population (204 lakhs), Hyderabad as the city with the lowest population (108 lakhs), and calculates the average population as 146.5 lakhs.

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```
[14]: print("CITY POPULATION ANALYSIS")
      print("-" * 35)

      print("Total Cities:", len(df))

      print(
          "Highest Population:",
          df.loc[df["Population (Lakhs)"].idxmax(), "City"]
      )

      print(
          "Lowest Population:",
          df.loc[df["Population (Lakhs)"].idxmin(), "City"]
      )

      print(
          "Average Population:",
          round(df["Population (Lakhs)"].mean(), 2),
          "Lakhs"
      )

CITY POPULATION ANALYSIS
-----
Total Cities: 6
Highest Population: Mumbai
Lowest Population: Hyderabad
Average Population: 146.5 Lakhs

[15]: fig.write_html(
      "city_population_chart.html"
      )

      print("Interactive chart saved successfully.")

Interactive chart saved successfully.

[16]: df.to_csv(
      "city_population.csv",
      index=False
      )

      print("Dataset saved successfully.")

Dataset saved successfully.

[ ]:
```

Fig. 1.6: Final Analysis Summary and File Saving Output

Explanation/Observation: The final summary confirms 6 cities were analyzed, with Mumbai highest and Hyderabad lowest. The interactive chart and CSV dataset are saved successfully.

Conclusions

The experiment was successfully completed using Pandas and Plotly. The population data was organized in a DataFrame, visualized with interactive bar charts, and sorted for comparison. Mumbai had the highest population, Hyderabad the lowest, and the average population was 146.5 lakhs. The interactive chart was saved as HTML and the dataset as CSV.

References

1. Michael Freeman and Joel Ross, Programming Skills for Data Science, Addison-Wesley, 2018.
2. Python Program (GitHub link): <https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%206/Week-6-PP.ipynb>
3. Class lecture link: [Open Lecture File](#)

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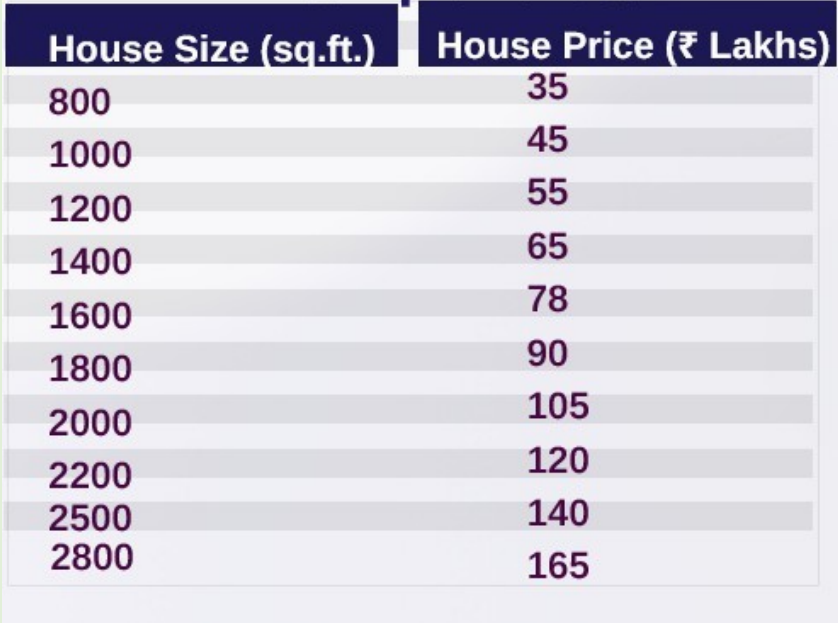
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Program S. No. 12

Title	Interactive Scatter Plot with Plotly			
Problem Statement	Create an interactive scatter plot showing the relationship between house size and price.			
Objectives	• To understand interactive scatter plot visualization using Plotly. • To organize house size and house price data using a Pandas DataFrame. • To visualize and interpret the relationship between house size and house price. • To compute descriptive measures such as minimum, maximum, average, and correlation. • To create a trendline for the relationship and save the interactive chart and dataset for further use.			
Dataset URL Link / File	 Dataset source: Manually defined in the Jupyter Notebook using the house-size and house-price values shown above; no external dataset URL is required.			
Functions Details	Sl. No.	Function	Package	Description of function
	1	pd.DataFrame()	pandas	Creates a structured DataFrame from the house size and price data.
	2	px.scatter()	plotly.express	Creates an interactive scatter plot from the DataFrame.
	3	update_traces()	plotly	Updates marker size and text-position properties of the plotted points.

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	4	update_layout()	plotly	Updates the chart layout and axis titles.
	5	show()	plotly	Displays the interactive Plotly chart inside the Jupyter Notebook.
	6	sort_values()	pandas	Sorts the dataset based on house size in ascending order.
	7	idxmax()	pandas	Returns the index of the record with the maximum house price.
	8	idxmin()	pandas	Returns the index of the record with the minimum house price.
	9	mean()	pandas	Calculates the average house price and average house size.
	10	corr()	pandas	Calculates the correlation between house size and house price.
	11	write_html()	plotly	Saves the interactive Plotly scatter plot as an HTML file.
	12	to_csv()	pandas	Saves the house-price DataFrame as a CSV file without the index column.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model		1. Import Pandas and Plotly Express for tabular data handling and interactive visualization. 2. Create a dictionary containing house sizes in square feet and corresponding house prices in lakhs. 3. Convert the dictionary into a Pandas DataFrame and display all ten records. 4. Verify the dataset shape and print the complete house-size and house-price lists. 5. Create a basic interactive scatter plot with house size on the X-axis and house price on the Y-axis. 6. Create an improved scatter plot with larger markers, value labels, title, and proper axis labels. 7. Sort the DataFrame by house size in ascending order and display the sorted dataset. 8. Use idxmax() and idxmin() to identify the most expensive and least expensive houses. 9. Calculate the average house price and average house size using mean(). 10. Calculate the correlation between house size and price using corr() to measure the strength of their relationship. 11. Create a scatter plot with an OLS trendline to visually represent the positive relationship between size and price.		

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	12. Display the final house-price analysis and save the interactive chart as HTML and the dataset as CSV. Flow: START → Import Libraries → Create Dataset → DataFrame → Verify Data → Scatter Plot → Enhanced Plot → Sort Data → Price Statistics → Average Values → Correlation → Trendline → Save HTML/CSV → STOP
Program	<pre>import pandas as pd import plotly.express as px data = { "House Size (sq.ft)": [800, 1000, 1200, 1400, 1600, 1800, 2000, 2200, 2500, 2800], "House Price (Lakhs)": [35, 45, 55, 65, 78, 90, 105, 120, 140, 165] } df = pd.DataFrame(data) df print("Dataset Shape:", df.shape) print("\nHouse Sizes:") print(df["House Size (sq.ft.)"].tolist()) print("\nHouse Prices:") print(df["House Price (Lakhs)"].tolist())</pre>

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```
fig = px.scatter(
    df,
    x="House Size (sq.ft.)",
    y="House Price (Lakhs)",
    title="House Size vs House Price"
)
fig.show()

fig = px.scatter(
    df,
    x="House Size (sq.ft.)",
    y="House Price (Lakhs)",
    title="Relationship Between House Size and Price",
    text="House Price (Lakhs)"
)
fig.update_traces(
    marker=dict(size=12),
    textposition="top center"
)
fig.update_layout(
    xaxis_title="House Size (sq.ft.)",
    yaxis_title="House Price (Lakhs)"
)
fig.show()

sorted_df = df.sort_values(
    by="House Size (sq.ft.)",
    ascending=True
)
sorted_df

highest = df.loc[
    df["House Price (Lakhs)"].idxmax()
]
print("Most Expensive House:")
print(highest)

lowest = df.loc[
    df["House Price (Lakhs)"].idxmin()
]
print(
    "Least Expensive House:",
    lowest["House Size (sq.ft.)"],
    "sq.ft. - ",
    lowest["House Price (Lakhs)"],
```

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```
"Lakhs"
)

average_price = df["House Price (Lakhs)"].mean()
print(
    "Average House Price:",
    round(average_price, 2),
    "Lakhs"
)

average_size = df["House Size (sq.ft.)"].mean()
print(
    "Average House Size:",
    round(average_size, 2),
    "sq.ft."
)

correlation = df[
    "House Size (sq.ft.)"
].corr(
    df["House Price (Lakhs)"]
)
print(
    "Correlation between House Size and Price:",
    round(correlation, 3)
)

fig = px.scatter(
    df,
    x="House Size (sq.ft.)",
    y="House Price (Lakhs)",
    title="House Size vs Price with Trendline",
    trendline="ols"
)
fig.update_traces(
    marker=dict(size=12)
)
fig.update_layout(
    xaxis_title="House Size (sq.ft.)",
    yaxis_title="House Price (Lakhs)"
)
fig.show()

print("HOUSE PRICE ANALYSIS")
print("-" * 40)
print("Total Houses:", len(df))
```

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	<pre> print("Smallest House:", df["House Size (sq.ft.)"].min(), "sq.ft.") print("Largest House:", df["House Size (sq.ft.)"].max(), "sq.ft.") print("Lowest Price: ", df["House Price (Lakhs)"].min(), "Lakhs") print("Highest Price: ", df["House Price (Lakhs)"].max(), "Lakhs") print("Average Price: ", round(df["House Price (Lakhs)"].mean(), 2), "Lakhs") print("Correlation:", round(correlation, 3)) fig.write_html("house_size_price_scatter.html") print("Interactive scatter plot saved successfully.") df.to_csv("house_price_data.csv", index=False) print("Dataset saved successfully.") </pre>
Output Snapshots With Explanation/ Observations	The following snapshots are taken from the completed Jupyter Notebook. The images retain the complete visible code/output blocks from each source page; only unused white margins were removed so no program output is cut off.

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```
165
]
}

[3]: df = pd.DataFrame(data)

df

[3]:   House Size (sq.ft.)  House Price (Lakhs)
0                800             35
1               1000             45
2               1200             55
3               1400             65
4               1600             78
5               1800             90
6               2000            105
7               2200            120
8               2500            140
9               2800            165

[4]: print("Dataset Shape:", df.shape)

print("\nHouse Sizes:")
print(df["House Size (sq.ft.)"].tolist())

print("\nHouse Prices:")
print(df["House Price (Lakhs)"].tolist())

Dataset Shape: (10, 2)

House Sizes:
[800, 1000, 1200, 1400, 1600, 1800, 2000, 2200, 2500, 2800]

House Prices:
[35, 45, 55, 65, 78, 90, 105, 120, 140, 165]

[5]: fig = px.scatter(
    df,
    x="House Size (sq.ft.)",
    y="House Price (Lakhs)",
    title="House Size vs House Price"
)

fig.show()
```

2

Fig. 1.1: DataFrame Creation, Dataset Verification, and Basic Scatter Plot Code

Explanation/Observation: The dataset is converted into a Pandas DataFrame with 10 rows and 2 columns. The complete house-size and house-price lists are verified, and the code for the initial interactive scatter plot is executed.

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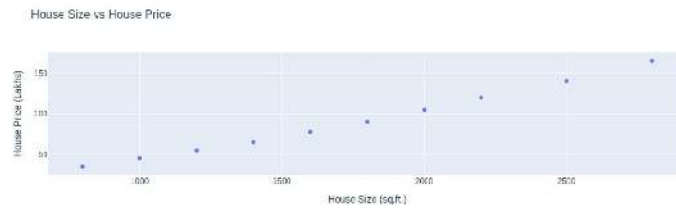
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```
[6]: fig = px.scatter(  
    df,  
    x="House Size (sq.ft.)",  
    y="House Price (Lakhs)",  
    title="Relationship Between House Size and Price",  
    text="House Price (Lakhs)"  
)  
  
fig.update_traces(  
    marker=dict(size=12),  
    textposition="top center"  
)  
  
fig.update_layout(  
    xaxis_title="House Size (sq.ft.)",  
    yaxis_title="House Price ( Lakhs)"  
)  
  
fig.show()
```



3

Fig. 1.2: Interactive and Enhanced Scatter Plots

Explanation/Observation: The first scatter plot shows house size versus price. The enhanced Plotly chart adds larger markers and price labels, making the strong upward relationship between size and price visually clear.

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```
[7]: sorted_df = df.sort_values(
      by="House Size (sq.ft.)",
      ascending=True
    )

sorted_df

[7]:   House Size (sq.ft.)  House Price (Lakhs)
0              800             35
1             1000             45
2             1200             55
3             1400             65
4             1600             78
5             1800             90
6             2000            105
7             2200            120
8             2500            140
9             2800            165

[8]: highest = df.loc[
      df["House Price (Lakhs)"].idxmax()
    ]

print("Most Expensive House:")
print(highest)

Most Expensive House:
House Size (sq.ft.)    2800
House Price (Lakhs)    165
Name: 9, dtype: int64

[9]: lowest = df.loc[
      df["House Price (Lakhs)"].idxmin()
    ]

print(
    "Least Expensive House:",
    lowest["House Size (sq.ft.)"],
    "sq.ft. - ",
    lowest["House Price (Lakhs)"],
    "Lakhs"
)

Least Expensive House: 800 sq.ft. - 35 Lakhs

[10]: average_price = df["House Price (Lakhs)"].mean()

print()
```

4

Fig. 1.3: Sorted Dataset and Extreme Price Analysis

Explanation/Observation: The dataset is sorted by house size in ascending order. The program identifies the 2800 sq.ft. house priced at 165 lakhs as the most expensive and the 800 sq.ft. house priced at 35 lakhs as the least expensive.

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```
"Average House Price:",
round(average_price, 2),
"Lakhs"
)

Average House Price: 89.8 Lakhs

[11]: average_size = df["House Size (sq.ft.)"].mean()

print(
    "Average House Size:",
    round(average_size, 2),
    "sq.ft."
)

Average House Size: 1730.0 sq.ft.

[12]: correlation = df[
    "House Size (sq.ft.)"
].corr(
    df["House Price (Lakhs)"]
)

print(
    "Correlation between House Size and Price:",
    round(correlation, 3)
)

Correlation between House Size and Price: 0.996

[14]: fig = px.scatter(
    df,
    x="House Size (sq.ft.)",
    y="House Price (Lakhs)",
    title="House Size vs Price with Trendline",
    trendline="ols"
)

fig.update_traces(
    marker=dict(size=12)
)

fig.update_layout(
    xaxis_title="House Size (sq.ft.)",
    yaxis_title="House Price ( Lakhs)"
)

fig.show()
```

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Fig. 1.4: Average Values, Correlation, and Trendline Program

Explanation/Observation: The average house price is 89.8 lakhs and the average house size is 1730 sq.ft. The calculated correlation is 0.996, indicating a very strong positive relationship. The trendline scatter-plot code is also shown.

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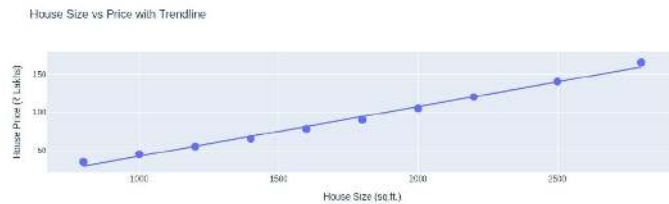
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```
[15]: print("HOUSE PRICE ANALYSIS")
      print("-" * 40)

      print("Total Houses:", len(df))

      print(
          "Smallest House:",
          df["House Size (sq.ft.)"].min(),
          "sq.ft."
      )

      print(
          "Largest House:",
          df["House Size (sq.ft.)"].max(),
          "sq.ft."
      )

      print(
          "Lowest Price: ",
          df["House Price (Lakhs)"].min(),
          "Lakhs"
      )

      print(
          "Highest Price: ",
          df["House Price (Lakhs)"].max(),
          "Lakhs"
      )

      print(
          "Average Price: ",
          round(df["House Price (Lakhs)"].mean(), 2),
          "Lakhs"
```

6

Fig. 1.5: Scatter Plot with OLS Trendline and Final Analysis Code

Explanation/Observation: The Plotly scatter plot with an OLS trendline confirms that house price rises strongly with house size. The final analysis code computes the smallest/largest houses and lowest/highest/average prices.

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```

)

print(
    "Correlation:",
    round(correlation, 3)
)

HOUSE PRICE ANALYSIS
-----

Total Houses: 10
Smallest House: 800 sq.ft.
Largest House: 2800 sq.ft.
Lowest Price: 35 Lakhs
Highest Price: 165 Lakhs
Average Price: 89.8 Lakhs
Correlation: 0.996

[16]: fig.write_html(
        "house_size_price_scatter.html"
    )

    print("Interactive scatter plot saved successfully.")

Interactive scatter plot saved successfully.

[17]: df.to_csv(
        "house_price_data.csv",
        index=False
    )

    print("Dataset saved successfully.")

Dataset saved successfully.

[ ]:

```

7

Fig. 1.6: Final House Price Analysis and File Saving Output

Explanation/Observation: The final output confirms 10 houses, sizes from 800 to 2800 sq.ft., prices from 35 to 165 lakhs, an average price of 89.8 lakhs, and correlation of 0.996. The interactive HTML chart and CSV dataset are saved successfully.

Conclusions

The experiment was successfully completed using Pandas and Plotly. House size and price data were organized in a Pandas DataFrame and visualized through interactive scatter plots. The analysis found an average house size of 1730 sq.ft., an average price of 89.8 lakhs, and a correlation of 0.996, showing a very strong positive relationship between house size and price. An OLS trendline further confirmed this relationship. The interactive

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	chart and dataset were successfully saved as HTML and CSV files.
References	1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018. 2. Python Program (GitHub link): https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%206/Week-6-PP2.ipynb 3. Class lecture link: Open Lecture File

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Program S. No. 13

Title	Interactive Scatter Plot with Bokeh			
Problem Statement	Create an interactive scatter plot showing the relationship between petal length and petal width from the Iris dataset using Bokeh.			
Objectives	• To understand interactive scatter plot visualization using Bokeh. • To load and organize the Iris dataset using Pandas and scikit-learn. • To visualize the relationship between petal length and petal width. • To differentiate Iris species using separate colors and an interactive legend. • To use Bokeh hover, zoom, pan, reset and save tools for interactive analysis.			
Dataset URL Link / File	Dataset: Built-in Iris dataset loaded using sklearn.datasets.load_iris(). Selected attributes: Petal Length (cm), Petal Width (cm), and Species.			
	Species		Samples	
	setosa		50	
	versicolor		50	
	virginica		50	
Functions Details	Sl. No.	Function	Package	Description of function
	1	load_iris()	sklearn	Loads the built-in Iris dataset.
	2	rename()	Pandas	Renames dataset columns for clearer labels.
	3	map()	Pandas	Maps numeric target values to species names.
	4	DataFrame()	Pandas	Represents the selected Iris attributes in tabular form.
	5	ColumnData Source()	Bokeh	Creates a Bokeh-friendly data source.
	6	figure()	Bokeh	Creates the interactive plotting canvas.
	7	scatter()	Bokeh	Plots petal length against petal width.
	8	HoverTool()	Bokeh	Displays species and measurements on hover.
	9	add_tools()	Bokeh	Adds the hover tool to the Bokeh figure.
	10	value_counts()	Pandas	Counts the records belonging to each Iris species.
	11	corr()	Pandas	Calculates correlation between petal length and width.
	12	output_file()/save()	Bokeh	Saves the interactive visualization as an HTML file.

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Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Pandas, the Iris dataset utility, and required Bokeh modules. 2. Load the Iris dataset and rename the petal measurement columns for clarity. 3. Convert numeric target values into the species names setosa, versicolor and virginica. 4. Create a DataFrame containing Petal Length, Petal Width and Species. 5. Display sample records, dataset shape, species counts and correlation for verification. 6. Create a Bokeh figure with interactive pan, zoom, reset and save tools. 7. Plot one scatter series for each Iris species using a ColumnDataSource. 8. Add HoverTool so that species and measurements appear when the pointer is placed over a point. 9. Enable an interactive legend so species can be hidden or displayed. 10. Save the interactive Bokeh chart as an HTML file and verify successful execution. Flow: START -> Load Iris Dataset -> Prepare DataFrame -> ColumnDataSource -> Bokeh Figure -> Scatter Plot -> Hover/Legend -> Save HTML -> STOP
Program	<pre>import pandas as pd from sklearn.datasets import load_iris from bokeh.plotting import figure, output_file, save from bokeh.models import ColumnDataSource, HoverTool from bokeh.palettes import Category10 # Load the Iris dataset iris = load_iris(as_frame=True) df = iris.frame.rename(columns={ "petal length (cm)": "Petal Length (cm)", "petal width (cm)": "Petal Width (cm)" }) df["Species"] = df["target"].map(dict(enumerate(iris.target_names))) df = df[["Petal Length (cm)", "Petal Width (cm)", "Species"]] print(df.head(10)) print("Dataset Shape:", df.shape) print("Species Counts:") print(df["Species"].value_counts()) correlation = df["Petal Length (cm)"].corr(df["Petal Width (cm)"]) print("Correlation:", round(correlation, 3)) # Create the interactive Bokeh figure p = figure(title="Interactive Scatter Plot: Iris Petal Length vs Petal Width", x_axis_label="Petal Length (cm)", y_axis_label="Petal Width (cm)",</pre>

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```
width=900,
height=520,
tools="pan,wheel_zoom,box_zoom,reset,save"
)

colors = Category10[3]
for color, species in zip(colors, iris.target_names):
    subset = df[df["Species"] == species]
    source = ColumnDataSource(subset)
    p.scatter(
        x="Petal Length (cm)",
        y="Petal Width (cm)",
        source=source,
        size=9,
        color=color,
        alpha=0.75,
        legend_label=species
    )

hover = HoverTool(tooltips=[
    ("Species", "@Species"),
    ("Petal Length", "@{Petal Length (cm)}{0.0} cm"),
    ("Petal Width", "@{Petal Width (cm)}{0.0} cm")
])
p.add_tools(hover)
p.legend.title = "Species"
p.legend.location = "top_left"
p.legend.click_policy = "hide"

output_file("program13_iris_scatter.html")
save(p)
print("Interactive scatter plot saved successfully.")
```

**Output
Snapshots
With
Explanation/
Observations**

The following outputs were generated by executing the complete Program 13 code. The static chart image represents the interactive Bokeh visualization; the HTML file retains hover, zoom, pan and reset functionality.

Jupyter Output - Iris Dataset Summary

```
First 10 rows: Petal Length (cm), Petal Width (cm), Species
1.4  0.2  setosa
1.4  0.2  setosa
1.3  0.2  setosa
1.5  0.2  setosa
1.4  0.2  setosa
...
Dataset Shape: (150, 3)
Species Counts: setosa 50 | versicolor 50 | virginica 50
Correlation (Petal Length vs Petal Width): 0.963
```

Fig. 1.1: Iris Dataset and Correlation Summary

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The output verifies that 150 Iris samples were loaded, with 50 records for each species. The petal length-petal width correlation is approximately 0.963, indicating a strong positive relationship.

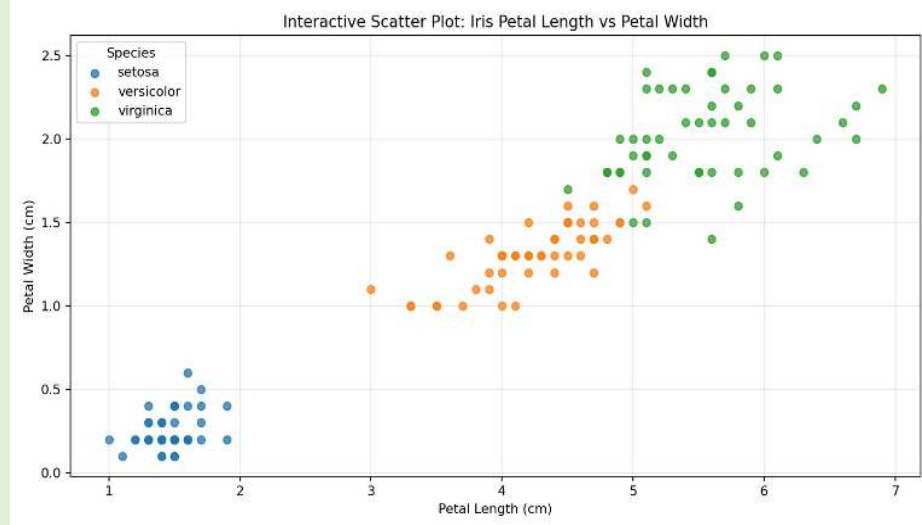


Fig. 1.2: Interactive Scatter Plot Output

The scatter plot shows three clearly distinguishable Iris species. Setosa forms a compact cluster at lower petal measurements, while versicolor and virginica occupy progressively larger petal-length and petal-width ranges.

Jupyter Output - Program 13 Final Status

```
Interactive scatter plot created successfully using Bokeh.
HTML file saved: program13_iris_scatter.html
Total records visualized: 150
Species displayed: setosa, versicolor, virginica
Hover, zoom, pan, reset and save tools are enabled.
```

Fig. 1.3: Program 13 Execution and HTML Save Confirmation

The final execution confirms successful generation of the interactive Bokeh scatter plot and its HTML output. Hover, zoom, pan, reset and save tools are available in the saved interactive file.

Conclusions

The experiment was completed successfully using Pandas, scikit-learn and Bokeh. The Iris dataset was loaded and organized into a DataFrame containing petal length, petal width and species. The interactive scatter plot clearly demonstrated a strong positive relationship between petal length and petal width and visually separated the three Iris species. Bokeh interaction tools further enabled hovering, zooming, panning and legend-based filtering, making the visualization suitable for exploratory analysis.

References

1. Bokeh Documentation: [Open Link](#)
2. Pandas Documentation: [Open Link](#)
3. scikit-learn Iris Dataset Documentation: [Open Link](#)
4. Python Program (GitHub link): [Open Week 7 Notebook](#)
5. Class lecture link: [Open Lecture Link](#)

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Program S. No. 14

Title	Interactive Line Plot with Bokeh			
Problem Statement	Create an interactive line plot showing the daily temperatures over a week using Bokeh.			
Objectives	• To understand interactive line visualization using Bokeh. • To organize daily temperature values in a Pandas DataFrame. • To visualize day-wise temperature changes over one week. • To identify the highest, lowest and average weekly temperature. • To use hover, zoom, pan, reset and save tools for interactive time-series analysis.			
Dataset URL Link / File	Dataset: Manually defined weekly temperature readings for seven consecutive days.			
	Day		Temperature (°C)	
	Monday		29	
	Tuesday		31	
	Wednesday		30	
	Thursday		32	
	Friday		34	
	Saturday		33	
	Sunday		31	
Functions Details	Sl. No.	Function	Package	Description of function
	1	DataFrame()	Pandas	Creates a tabular dataset from the day and temperature values.
	2	ColumnData Source()	Bokeh	Converts the DataFrame into a Bokeh data source.
	3	figure()	Bokeh	Creates the interactive Bokeh plotting canvas.
	4	line()	Bokeh	Draws the line connecting daily temperatures.
	5	scatter()	Bokeh	Adds visible markers for each day.
	6	HoverTool()	Bokeh	Displays day and temperature values on hover.
	7	add_tools()	Bokeh	Adds the hover tool to the figure.
	8	idxmax()	Pandas	Locates the day with the maximum temperature.
	9	idxmin()	Pandas	Locates the day with the minimum temperature.
	10	mean()	Pandas	Calculates the average weekly temperature.
	11	output_file()	Bokeh	Defines the HTML output filename.
	12	save()	Bokeh	Saves the interactive line plot as an HTML file.

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Program S. No. 14

Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Pandas and the required Bokeh modules. 2. Define the seven days of the week and their corresponding temperature values. 3. Create a Pandas DataFrame and display it for verification. 4. Calculate the highest, lowest and average temperature values. 5. Convert the DataFrame into a Bokeh ColumnDataSource. 6. Create a Bokeh figure using the days as categorical X-axis values. 7. Plot the temperature trend using a line and add markers for each day. 8. Add HoverTool to display the exact day and temperature interactively. 9. Save the completed visualization as an HTML file and verify successful execution. Flow: START -> Create Weekly Dataset -> DataFrame -> Temperature Statistics -> ColumnDataSource -> Bokeh Figure -> Line + Markers -> Hover -> Save HTML -> STOP
Program	<pre>import pandas as pd from bokeh.plotting import figure, output_file, save from bokeh.models import ColumnDataSource, HoverTool # Weekly temperature dataset days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"] temperatures = [29, 31, 30, 32, 34, 33, 31] df = pd.DataFrame({"Day": days, "Temperature (°C)": temperatures}) print(df) highest_row = df.loc[df["Temperature (°C)"].idxmax()] lowest_row = df.loc[df["Temperature (°C)"].idxmin()] average_temp = df["Temperature (°C)"].mean() print("Highest Temperature:", highest_row["Day"], "-", highest_row["Temperature (°C)"], "°C") print("Lowest Temperature:", lowest_row["Day"], "-", lowest_row["Temperature (°C)"], "°C") print("Average Temperature:", round(average_temp, 2), "°C") source = ColumnDataSource(df) p = figure(x_range=days, title="Interactive Line Plot: Daily Temperature Over a Week", x_axis_label="Day", y_axis_label="Temperature (°C)", width=900, height=520, tools="pan,wheel_zoom,box_zoom,reset,save")</pre>

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```
p.line(x="Day", y="Temperature (°C)", source=source, line_width=3)
p.scatter(x="Day", y="Temperature (°C)", source=source, size=10)
```

```
hover = HoverTool(tooltips=[
    ("Day", "@Day"),
    ("Temperature", "@{Temperature (°C)} °C")
])
p.add_tools(hover)
p.xgrid.grid_line_color = None
```

```
output_file("program14_temperature_line.html")
save(p)
print("Interactive line plot saved successfully.")
```

**Output
Snapshots
With
Explanation/
Observations**

The following outputs were generated from the completed Program 14. The chart image is a static view of the interactive Bokeh line plot, while the saved HTML output contains the full interactive tools.

Jupyter Output - Weekly Temperature Summary

Weekly Temperature Dataset:
Monday 29 °C | Tuesday 31 °C | Wednesday 30 °C
Thursday 32 °C | Friday 34 °C | Saturday 33 °C | Sunday 31 °C
Highest Temperature: Friday - 34 °C
Lowest Temperature: Monday - 29 °C
Average Temperature: 31.43 °C

Fig. 2.1: Weekly Temperature Dataset and Statistical Summary

The dataset contains seven daily readings. Friday records the highest temperature of 34 °C, Monday records the lowest temperature of 29 °C, and the weekly average is approximately 31.43 °C.

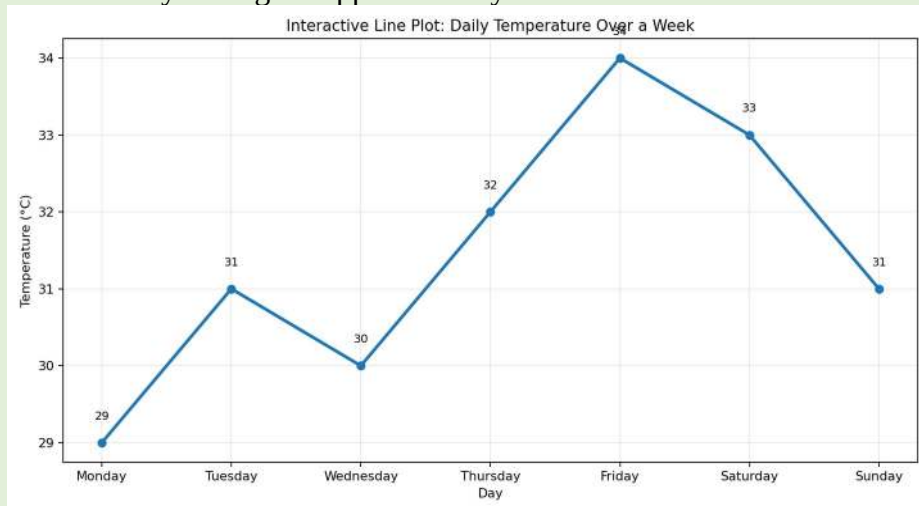


Fig. 2.2: Interactive Line Plot Output

The line plot shows the day-wise temperature variation across the week. The temperature rises overall from Monday to Friday, reaches its maximum on Friday, and decreases during the weekend.

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	Jupyter Output - Program 14 Final Status Interactive line plot created successfully using Bokeh. HTML file saved: program14_temperature_line.html Days visualized: 7 Hover, zoom, pan, reset and save tools are enabled. Fig. 2.3: Program 14 Execution and HTML Save Confirmation The execution confirms successful creation and saving of the Bokeh line plot. The resulting HTML file supports hover, zoom, pan, reset and save interactions.
Conclusions	The exercise was completed successfully using Pandas and Bokeh. Daily temperature values for one week were organized in a DataFrame and visualized using an interactive line plot. Statistical analysis identified Friday as the hottest day, Monday as the coolest day, and the average weekly temperature as approximately 31.43 °C. The interactive Bokeh output makes the weekly trend easy to inspect using hover and navigation tools.
References	1. Bokeh Documentation: Open Link 2. Pandas Documentation: Open Link 3. Python Program (GitHub link): Open Week 7 Notebook 4. Class lecture link: Open Lecture Link

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Program S. No. 15

Title	Interactive Bar Chart with Bokeh			
Problem Statement	Create an interactive bar chart showing the sales figures of different products using Bokeh.			
Objectives	• To understand interactive categorical bar-chart visualization using Bokeh. • To organize product-wise sales values in a Pandas DataFrame. • To compare sales performance across different products. • To calculate total, highest and average product sales. • To use Bokeh hover, zoom, pan, reset and save tools for interactive analysis.			
Dataset URL Link / File	Dataset: Manually defined product-wise sales values for six products.			
	Product		Sales (Units)	
	Laptop		125	
	Smartphone		210	
	Tablet		95	
	Headphones		170	
	Smartwatch		140	
	Camera		80	
Functions Details	Sl. No.	Function	Package	Description of function
	1	DataFrame()	Pandas	Creates a structured product-sales DataFrame.
	2	ColumnData Source()	Bokeh	Converts the DataFrame into a Bokeh data source.
	3	figure()	Bokeh	Creates the categorical plotting canvas.
	4	vbar()	Bokeh	Draws one vertical bar for each product.
	5	HoverTool()	Bokeh	Displays product and sales values on hover.
	6	add_tools()	Bokeh	Adds the hover tool to the figure.
	7	idxmax()	Pandas	Locates the product with the highest sales.
	8	max()	Pandas	Returns the highest sales value.
	9	sum()	Pandas	Calculates the total number of units sold.
	10	mean()	Pandas	Calculates the average sales per product.
	11	output_file()	Bokeh	Defines the interactive HTML output filename.
	12	save()	Bokeh	Saves the completed Bokeh bar chart as HTML.

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Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Pandas and the required Bokeh modules. 2. Define the product names and their corresponding sales figures. 3. Create and display a Pandas DataFrame for verification. 4. Calculate total sales, the highest-selling product and average sales. 5. Convert the DataFrame to a Bokeh ColumnDataSource. 6. Create a categorical Bokeh figure with product names on the X-axis. 7. Use vbar() to display the product-wise sales values. 8. Add HoverTool so exact sales values are visible interactively. 9. Save the completed interactive bar chart as an HTML file and verify successful execution. Flow: START -> Create Product Dataset -> DataFrame -> Sales Statistics -> ColumnDataSource -> Bokeh Figure -> Vertical Bars -> Hover -> Save HTML -> STOP
Program	<pre>import pandas as pd from bokeh.plotting import figure, output_file, save from bokeh.models import ColumnDataSource, HoverTool # Product sales dataset products = ["Laptop", "Smartphone", "Tablet", "Headphones", "Smartwatch", "Camera"] sales = [125, 210, 95, 170, 140, 80] df = pd.DataFrame({"Product": products, "Sales (Units)": sales}) print(df) highest_row = df.loc[df["Sales (Units)"].idxmax()] total_sales = df["Sales (Units)"].sum() average_sales = df["Sales (Units)"].mean() print("Total Sales:", total_sales, "units") print("Highest Sales:", highest_row["Product"], "-", highest_row["Sales (Units)"], "units") print("Average Sales:", round(average_sales, 2), "units") source = ColumnDataSource(df) p = figure(x_range=products, title="Interactive Bar Chart: Product Sales", x_axis_label="Product", y_axis_label="Sales (Units)", width=900, height=520, tools="pan,wheel_zoom,box_zoom,reset,save") p.vbar(x="Product", top="Sales (Units)", source=source, width=0.65)</pre>

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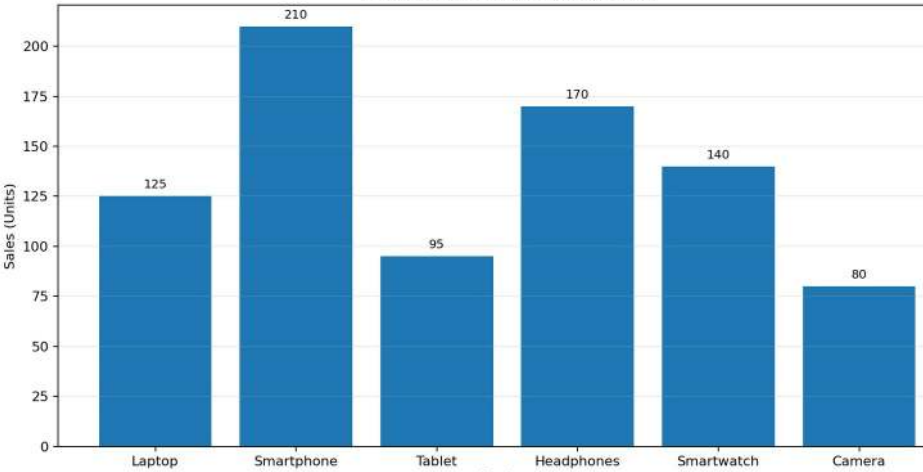
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	<pre> hover = HoverTool(tooltips=[("Product", "@Product"), ("Sales", "@{Sales (Units)} units")]) p.add_tools(hover) p.xgrid.grid_line_color = None output_file("program15_product_sales_bar.html") save(p) print("Interactive bar chart saved successfully.") </pre>														
Output Snapshots With Explanation/ Observations	The following outputs were generated by executing Program 15. The static chart image presents the product-wise sales comparison, while the generated Bokeh HTML file preserves the interactive functionality. Jupyter Output - Product Sales Summary Product Sales Dataset: Laptop 125 Smartphone 210 Tablet 95 Headphones 170 Smartwatch 140 Camera 80 Total Sales: 820 units Highest Sales: Smartphone - 210 units Average Sales: 136.67 units Fig. 3.1: Product Sales Dataset and Statistical Summary The product dataset contains six sales values. Total sales are 820 units, Smartphone has the highest sales at 210 units, and average sales are approximately 136.67 units per product. Interactive Bar Chart: Product Sales  <table border="1" style="width: 100%; text-align: center; margin-top: 10px;"> <thead> <tr> <th>Product</th> <th>Sales (Units)</th> </tr> </thead> <tbody> <tr> <td>Laptop</td> <td>125</td> </tr> <tr> <td>Smartphone</td> <td>210</td> </tr> <tr> <td>Tablet</td> <td>95</td> </tr> <tr> <td>Headphones</td> <td>170</td> </tr> <tr> <td>Smartwatch</td> <td>140</td> </tr> <tr> <td>Camera</td> <td>80</td> </tr> </tbody> </table> Fig. 3.2: Interactive Bar Chart Output The bar chart compares sales across all six products. Smartphone is the highest-selling product, followed by Headphones, while Camera records the lowest sales.	Product	Sales (Units)	Laptop	125	Smartphone	210	Tablet	95	Headphones	170	Smartwatch	140	Camera	80
Product	Sales (Units)														
Laptop	125														
Smartphone	210														
Tablet	95														
Headphones	170														
Smartwatch	140														
Camera	80														

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	Jupyter Output - Program 15 Final Status Interactive bar chart created successfully using Bokeh. HTML file saved: program15_product_sales_bar.html Products visualized: 6 Hover, zoom, pan, reset and save tools are enabled. Fig. 3.3: Program 15 Execution and HTML Save Confirmation The final execution confirms successful generation and saving of the Bokeh product-sales bar chart. The HTML output supports hover and navigation tools for interactive inspection.
Conclusions	The exercise was completed successfully using Pandas and Bokeh. Product sales data were organized in a DataFrame and visualized through an interactive vertical bar chart. Smartphone recorded the highest sales with 210 units, total sales were 820 units, and average sales were approximately 136.67 units. The visualization provides a clear product-wise comparison and the interactive Bokeh tools improve data exploration.
References	1. Bokeh Documentation: Open Link 2. Pandas Documentation: Open Link 3. Python Program (GitHub link): Open Week 7 Notebook 4. Class lecture link: Open Lecture Link

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